SECURITY



Future-proof Security in TPM with SPDM and PQC Capabilities



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A Short View into the Past: Project Kirkland

- In 2022: *"TPM Transport Security: Defeating Active Interposers with DICE"*.*
  - Introduction of the concept that SPDM + DICE can solve TPM bus interposer attacks in data center environments.
  - TPM objects become bound to CPU/host identities.
  - Secure SPDM channel between host and TPM.
  - Remote verification model.



*OCP Global Summit 2022: TPM Transport Security: Defeating Active Interposers with DICE (Link)

What has Changed Since Project Kirkland?

Then:

- Protect TPM communication bus
- Transport security proposal
- Proof of feasibility



Now:

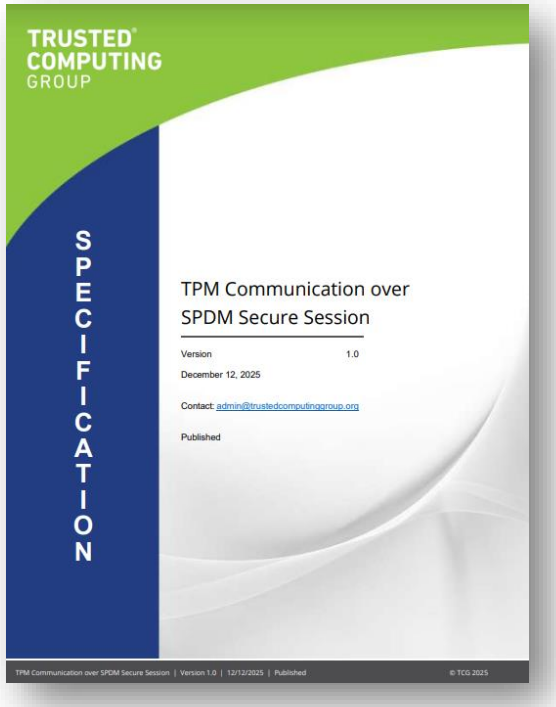
- TCG specification
- Standardized TPM over SPDM transport messages
- Policy integration
- Datacenter attestation guidance
- Industry support

TPM Becomes a Secured Endpoint

- Traditional TPM
  - If you trust the TPM, you trust the channel
- SPDM-enabled TPM
  - Authenticated session
  - Encrypted communication
  - Integrity protection
  - Session identity

- TPM learns **who** created the session and **which** firmware is running.
- Trust ***WHAT*** was measured **vs.** Trust ***WHO WAS MEASSURED*** and ***WHO COMMUNICATES.***
- By combining **TPM SPDM** + **DICE**, **active interposers attacks** are mitigated.

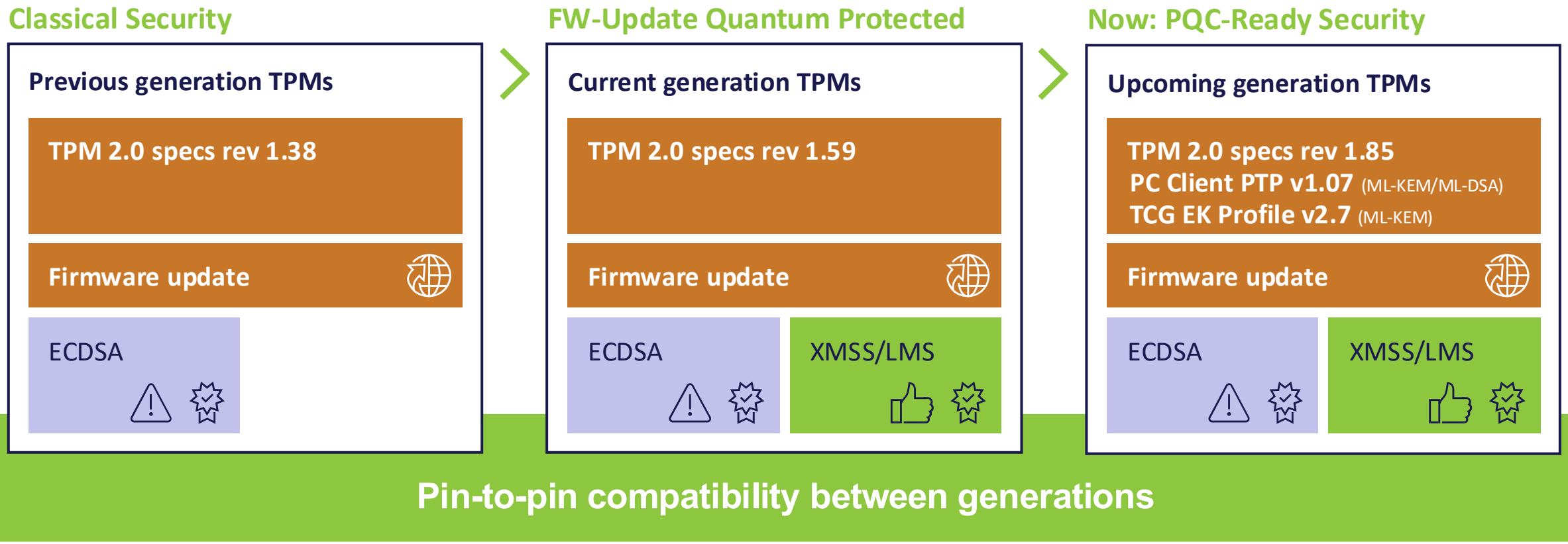
TCG TPM SPDM Specification



The SPDM specification defines a secure channel between host and TPM where TPM commands and responses are encapsulated inside SPDM secured messages.

Next challenge: Post-Quantum Cryptography

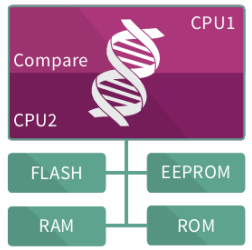
- PQC evolution in TPMs



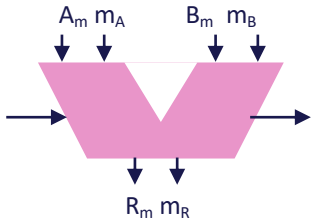
Building Blocks in TPMs to Answer PQC

- Security controller with protected memory for software-based PQC.
- Fault detection and mitigation to protect cryptographic operations.
- Hardware countermeasures against physical and side-channel attacks.

Secure processing foundation



PQC Accelerator



- Hardware support to speed up PQC related operations.
- An extended instruction set coprocessor for secured computation of masked data.
- Flexible to allow adoption to changed algorithms.

PQC

- PQC standards and implementation requirements will continue to evolve.
- This will require crypto adoptions in the years to come.
- Crypto agility by updating low layer is the best way to address this moving target.



Crypto agility – FW update



PQC crypto library

- A certified PQC cryptographic library can accelerate development and integration.
- In close interaction with HW.
- Allows software adjustments to protect against future attacks.

Demo!

- OPTIGA™ TPM SLB 9678-based PQC and SPDM Demonstrator

RASPBERRY PI

WEBSERVER
Python Flask

OPENSSL
v3.6.0-dev

TPM2-PYTSS
Added PQC Support

TPM2-TSS
Added PQC Support

Linux Kernel SPDM
Added SPDM Support

/dev/tpm0

To be released into the Linux Kernel



SPDM Channel

SPI

OPTIGA™ TPM SLB9678

TPM Firmware

TPM Commands

TPM2_Create

TPM2_Load

TPM2_SignDigest

ML-DSA
OBJECT

ML-KEM

...

IFX PQC CRYPTO LIBRARY

CC EAL6+ CERTIFIED

HashML-DSA-44	HashML-DSA-65	HashML-DSA-87
ML-DSA-44	ML-DSA-65	ML-DSA-87
ML-KEM-512	ML-KEM-768	ML-KEM-1024

SPDM Responder

Key Takeaway

1. SPDM protects the TPM channel through authenticated, encrypted, integrity-protected sessions.
2. DICE + SPDM bind trust to identities and measurements, helping mitigate active interposer attacks.
3. PQC and crypto agility future-proof TPM deployments as algorithms and standards continue to evolve.
4. Working demonstrator that shows the path forward with SPDM and PQC capabilities.

➢**TPMs** are evolving from isolated roots of trust into **attested**, **protocol-aware** security endpoints and **quantum-resistance** devices.

Call to Action

- **Stop by the Infineon Technologies booth E26 and see the live demo!**
- We have demonstrated that PQC and SPDM integration with TPMs is possible. Now we call on the wider OCP ecosystem to implement it.
- Start integrating PQC and SPDM into data center security architectures. Evaluate TPM capabilities and prototype hardware and software integration.
- Collaborate to accelerate adoption. Contribute to open-source implementations, share integration experience, and drive cross-vendor interoperability.

Thank You!

